# Zhengyang Yu

*Software Engineering Student*

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## Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

## Projects

2026 **Quotaquarium [App Store] [GitHub]**, *Personal Project*
- Shipped a production SwiftUI application for iPhone and Apple Watch to the App Store, owning requirements, implementation, testing, release compliance, and technical documentation.
- Designed reliable multi-component data flows with secure device authorization, Keychain, URLSession, caching, background refresh, structured diagnostics, and idempotent operations.
- Built a companion C++/FreeRTOS system with concurrent network workers, adaptive HTTPS polling, interactive rendering, and bounded persistent logs on an ESP32-S3.

2025 **HTML to ePub Converter**, *Personal Project*
- Built a Python data-processing pipeline that parses HTML and XML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved correctness and maintainability through modular transformations, explicit error handling, and repeatable output validation.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST workflow with explicit state transitions to support testing, debugging, and reliable smart-lighting behavior.

## Skills

Programming: Python, C++, Java, Swift, C
Systems & Data: Linux, REST APIs, HTTPS, JSON, XML, caching, concurrent and event-driven systems
Engineering: Algorithms, data structures, object-oriented design, design patterns, software testing, debugging
Delivery: Requirements analysis, technical documentation, App Store release, physical-device validation
AI Development: Codex-assisted codebase exploration, implementation, refactoring, debugging, and verification
Tools: Git, Docker, Linux, PlatformIO, Xcode, VS Code

## Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.